

WEEK-3

```
mysql> create database studentdb;  
Query OK, 1 row affected (0.02 sec)
```

```
mysql> use studentdb;  
ERROR 1049 (42000): Unknown database 'studentdb'  
mysql> use studentdb;  
Database changed  
mysql> CREATE TABLE roles(  
    -> role_id INT PRIMARY KEY,  
    -> role_name VARCHAR(50));  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> CREATE TABLE users(  
    -> user_id INT PRIMARY KEY,  
    -> name VARCHAR(50),  
    -> email VARCHAR(50),  
    -> role_id INT);  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE departments(  
    -> dept_id INT PRIMARY KEY,  
    -> dept_name VARCHAR(50));  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE employees(  
    -> emp_id INT PRIMARY KEY,  
    -> emp_name VARCHAR(50),  
    -> dept_id INT,  
    -> manager_id INT,  
    -> salary DECIMAL(10,2));  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE customers(  
    -> customer_id INT PRIMARY KEY,  
    -> customer_name VARCHAR(50),  
    -> city VARCHAR(50));  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE orders(  
    -> order_id INT PRIMARY KEY,  
    -> customer_id INT,  
    -> order_date DATE,  
    -> total_amount DECIMAL(10,2));  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE accounts(  
    -> account_no INT PRIMARY KEY,  
    -> holder_name VARCHAR(50),  
    -> balance DECIMAL(10,2));  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SELECT emp_name,dept_name FROM employees e INNER JOIN departments d ON e.dept_id=d.dept_id;  
Empty set (0.01 sec)
```

```
mysql> INSERT INTO roles VALUES  
    -> (1, 'Admin'),  
    -> (2, 'User'),  
    -> (3, 'Developer');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO users VALUES
-> (101, 'john', 'john@gmail.com', 1),
-> (102, 'Priya', 'priya@gmail.com', 2),
-> (103, 'Ravi', 'ravi@gmail.com', 3),
-> (104, 'Emma', 'emma@gmail.com', NULL);
```

Query OK, 4 rows affected (0.04 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO departments VALUES
-> (10, 'CSE'),
-> (20, 'ECE'),
-> (30, 'EEE'),
-> (40, 'MECH');
```

Query OK, 4 rows affected (0.03 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO employees VALUES
-> (1, 'Arun', 10, NULL, 80000),
-> (2, 'Bhavna', 10, 1, 60000),
-> (3, 'Charan', 20, 1, 45000),
-> (4, 'Divya', 30, 2, 55000),
-> (5, 'Esha', 10, 2, 70000),
-> (6, 'Farhan', 40, 4, 40000);
```

Query OK, 6 rows affected (0.01 sec)

Records: 6 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO customers VALUES
-> (1, 'Ramesh', 'Hyderabad'),
-> (2, 'Suresh', 'Vijayawada'),
-> (3, 'Anita', 'Warangal'),
-> (4, 'Kiran', 'Nizamabad');
```

Query OK, 4 rows affected (0.03 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO orders VALUES
-> (1001, 1, '2026-08-01', 1500.00),
-> (1002, 2, '2026-08-02', 2500.00),
-> (1003, 1, '2026-08-03', 3000.00),
-> (1004, 3, '2026-08-03', 1200.00),
-> (1005, 4, '2026-08-04', 4500.00);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO accounts VALUES
-> (101, 'Raj', 50000.00),
-> (102, 'Kiran', 30000.00),
-> (103, 'Meena', 45000.00);
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT emp_name,dept_name FROM employees e INNER JOIN departments
d ON e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Arun     | CSE       |
| Bhavna   | CSE       |
| Charan   | ECE       |
| Divya    | EEE       |
| Esha     | CSE       |
| Farhan   | MECH      |
+-----+-----+
6 rows in set (0.02 sec)
```

```
mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.ro
le_id=r.role_id;
+-----+-----+-----+
| name  | email          | role_name |
+-----+-----+-----+
| john  | john@gmail.com | Admin     |
| Priya | priya@gmail.com | User      |
| Ravi  | ravi@gmail.com  | Developer |
| Emma  | emma@gmail.com  | NULL      |
+-----+-----+-----+
4 rows in set (0.02 sec)
```

```
mysql> SELECT name,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
```

name	role_name
john	Admin
Priya	User
Ravi	Developer

```
3 rows in set (0.02 sec)
```

```
mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name,email,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
```

name	email	role_name
john	john@gmail.com	Admin
Priya	priya@gmail.com	User
Ravi	ravi@gmail.com	Developer
Emma	emma@gmail.com	NULL

```
4 rows in set (0.02 sec)
```

```
mysql> SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.emp_id;
```

Employee	Manager
Arun	NULL
Bhavna	Arun
Charan	Arun
Divya	Bhavna
Esha	Bhavna
Farhan	Divya

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT o.order_id,customer_name,city,total_amount FROM customers c JOIN orders o ON c.customer_id=o.customer_id;
```

order_id	customer_name	city	total_amount
1001	Ramesh	Hyderabad	1500.00
1002	Suresh	Vijayawada	2500.00
1003	Ramesh	Hyderabad	3000.00
1004	Anita	Warangal	1200.00
1005	Kiran	Nizamabad	4500.00

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e
JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;
+-----+-----+-----+-----+
| dept_name | COUNT(*) | AVG(salary) | MAX(salary) |
+-----+-----+-----+-----+
| CSE       | 3       | 70000.000000 | 80000.00    |
| ECE       | 1       | 45000.000000 | 45000.00    |
| EEE       | 1       | 55000.000000 | 55000.00    |
| MECH      | 1       | 40000.000000 | 40000.00    |
+-----+-----+-----+-----+
4 rows in set (0.02 sec)
```

```
mysql> SELECT emp_name,salary,dept_name FROM employees e JOIN departments
d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='CSE';
+-----+-----+-----+
| emp_name | salary | dept_name |
+-----+-----+-----+
| Arun     | 80000.00 | CSE       |
| Bhavna   | 60000.00 | CSE       |
| Esha     | 70000.00 | CSE       |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT customer_name,SUM(total_amount) TotalPurchase FROM customer
s c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name O
RDER BY TotalPurchase DESC;
+-----+-----+
| customer_name | TotalPurchase |
+-----+-----+
| Ramesh        | 4500.00       |
| Kiran         | 4500.00       |
| Suresh        | 2500.00       |
| Anita         | 1200.00       |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE faculty (
->     faculty_id INT PRIMARY KEY,
->     faculty_name VARCHAR(50)
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> INSERT INTO faculty VALUES
-> (1,'Dr. Rao'),
-> (2,'Dr. Sharma');
```

Query OK, 2 rows affected (0.02 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> CREATE TABLE courses (  
->     course_id INT PRIMARY KEY,  
->     course_name VARCHAR(50),  
->     faculty_id INT  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> CREATE TABLE students (  
->     student_id INT PRIMARY KEY,  
->     student_name VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> CREATE TABLE enrollments (  
->     student_id INT,  
->     course_id INT,  
->     PRIMARY KEY(student_id, course_id)  
-> );  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> INSERT INTO courses VALUES
-> (101,'DBSE',1),
-> (102,'Java',2);
Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO students VALUES
-> (1,'Akhil'),
-> (2,'Neha'),
-> (3,'Rohit');
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO enrollments VALUES
-> (1,101),
-> (1,102),
-> (2,101),
-> (3,102);
Query OK, 4 rows affected (0.03 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN
enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=
c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Akhil        | DBSE        | Dr. Rao      |
| Akhil        | Java        | Dr. Sharma   |
| Neha         | DBSE        | Dr. Rao      |
| Rohit        | Java        | Dr. Sharma   |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION; INSERT INTO accounts VALUES(104,'Anil',25000);
COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```



```
mysql> START TRANSACTION; INSERT INTO accounts VALUES(105,'Ravi',10000);  
ROLLBACK;  
Query OK, 0 rows affected (0.00 sec)  
  
Query OK, 1 row affected (0.00 sec)  
  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE  
account_no=101; UPDATE accounts SET balance=balance+5000 WHERE account_no  
=102; COMMIT;  
Query OK, 0 rows affected (0.00 sec)  
  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE  
account_no=101; UPDATE accounts SET balance=balance+3000 WHERE account_no  
=999; ROLLBACK;  
Query OK, 0 rows affected (0.00 sec)  
  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 0 Changed: 0 Warnings: 0  
  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance+2000 WHERE  
account_no=101; SAVEPOINT s1; UPDATE accounts SET balance=balance-1000 WH  
ERE account_no=101; ROLLBACK TO s1; COMMIT;  
Query OK, 0 rows affected (0.00 sec)  
  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
Query OK, 0 rows affected (0.00 sec)  
  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
Query OK, 0 rows affected (0.00 sec)  
  
Query OK, 0 rows affected (0.03 sec)
```